MACQUARIE University BUSINESS SCHOOL SYDNEY·AUSTRALIA

BUSA8090 Data and Visualisation for Business, Session 2, 2025

Assessment Task	Report - Visualisations
Due date	Week 13 – Wednesday 5th November 2025 11:55pm
Weight (%)	30%
Task description	This is an individual assessment - This assignment aims to enhance your data analysis and visualisation skills using Tableau or R. You will work with real-world data to uncover insights and present them through compelling visualisations.
Submission Method	On iLearn
Feedback mechanism(s)	On iLearn/ Via TurnItIn
Feedback available (anticipated date)	2-3 weeks post due date
Links to Unit Learning Outcomes	On successful completion you will be able to (ULO1, ULO4, ULO5): • Articulate the importance and application of data in a variety of contexts. • Examine and employ a variety of data visualisation techniques using R or Tableau. • Evaluate various popular data visualisation solutions.

INTEGRITY MATTERS

Integrity matters – at home, in your workplace and here at University. As a highly valued member of the Macquarie University Community, you carry great responsibility to uphold the good name of our institution. Our reputation is your reputation and will stay with you for life when your degree opens amazing opportunities for you.

If you are ever unsure whether your actions fall within the guidelines of Academic Integrity, please don't hesitate to reach out. Contact the Academic Literacies Unit or your Tutor/Unit Convenor.

ASSESSMENT DESCRIPTION

In this assessment, students will analyse real world crash data from Victoria roads using Tableau or R. The task involves creating insightful visualisations to uncover patterns related to crash trends, severity, contributing factors, and vehicle involvement. Students will also write a report interpreting their findings, discussing the analytical process, and proposing evidence based recommendations to

improve road safety. This assignment emphasises practical data visualisation skills and critical thinking using authentic government data.

Skills in focus for this assessment

- *Basic syntax knowledge of R for operations and functions.*
- *Familiarity of R packages and relevance to data manipulation and visualisation.*
- *Data importing.*
- *Understanding how to convert data types.*
- *Graphical representation skills for various types of visualisations.*
- Understanding the basic statistics for data analysis.
- Critical thinking for analysing data, identify patterns or trends and draw meaningful conclusions.
- Interpretation of results of data analysis and visualisations.
- Familiarity of R Studio development environment for R programming and visualisations.
- Familiarity of Tableau environment for visualisation.
- Navigating Tableau and leveraging its features (filters, maps, parameters)
- Interpreting real-world data from open government sources
- Selecting appropriate chart types for different data insights

ASSESSMENT INSTRUCTIONS

Dataset: Utilise the "Crash data from Victoria roads" dataset provided by Transport Victoria.

This dataset contains detailed information on road traffic crashes in Victoria, including variables such as date, location, crash severity, and contributing factors. The dataset is available in 3 CSV format files and can be downloaded from iLearn. The following are the files available:

vic_road_crash_data: A single flat file containing a subset of the attributes

Vehicle: The VEHICLE table contains the details of the vehicles involved with an accident. There can be one or more vehicles involved with one accident only. The accident_no and vehicle_id fields are the primary keys.

Person: The PERSON table contains the details of the people involved with an accident. There can be one or more people involved with one accident only. The accident_no and person_id fields are the primary keys.

Data Dictionary: Provides Supporting documentation of of the above files with further details of the attributes

Part A Practical Work (Total: 20 marks)

Create a series of 5 visualisations in Tableau or R to analyse and interpret the road crash data. Your visualisations must address the following:

1. Develop a time series visualisation to examine trends in road crashes over a specified period. Identify any patterns, trends or anomalies in the data.
2. Create a map visualisation to illustrate the geographic distribution of road crashes across different regions in Victoria. Highlight areas with higher incidences.

3. Design visualisations to explore the severity of crashes (e.g., serious injury, fatal, minor injury) and how severity levels vary across relevant factors such as location, time, or demographics.
4. Design visualisations to explore how crash severity varies across vehicle related factors such as year of manufacture, body style, make, model, and power. Consider whether particular vehicle types, ages, or specifications are associated with higher crashes or severity outcomes and present your findings clearly.

Part B: Write 1200–1500 words report to discuss the following: (Total: 10 marks)

1. Explain any data preparation, transformation and cleaning processes undertaken before analysis.
2. Present each visualisation developed in Part A, accompanied by an in-depth analysis of the findings and describe the rationale behind choosing specific visualisation types for each aspect of the analysis.
3. Interpret the patterns, trends, and anomalies observed in the visualisations and discuss the potential factors contributing to the observed data patterns.
4. Ethics plays a crucial role in data visualization. Discuss and apply three ethical principles you have adhered to, and explain how they are relevant to data visualisation practices in the industry

1. **Conclusion:**

   Summarise the key insights gained from the analysis.

Note: Structure your report with appropriate headings and subheadings. You may support your discussion with references to relevant literature, reports, and data sources.

ASSESSTIPS & FAQs

TIPS

Start Early: The dataset is large and may require cleaning, give yourself time to explore and understand it.

Have a Plan: Before diving into visualisations, outline your analysis plan. Understand the objectives, the data you're working with, and the steps you might need to take to reach your conclusions.

Understand Your Data: Spend time exploring and understanding your data. Use summary statistics and initial visualisations to get a feel for the data's structure, variables, and potential quirks.

Focus on Insights: Don't just create pretty charts. Ask: *What does this tell us about road safety in Victoria?*

Use a Variety of Visuals: Combine different types of charts (maps, line graphs, bar charts) to explore different angles of the data.

FAQs

Q: How do I choose the right type of visualisation?

A: The choice of visualisation depends on the nature of your data (categorical vs. numerical) and the story you want to tell.

Q: Can I use both Tableau and R in my assignment?

A: You can, but it's not required. You may choose **either** tool for all visualisations or combine them if it helps you tell a better story.

Q: How many visualizations do I need?

A: Five well-explained and relevant visualisations are expected, each offering a unique insight into the data.

Q: Can I use other packages beyond what was learned?

A: Absolutely! R has a vast ecosystem of packages. If you find one that suits your needs better, feel free to use it. Just ensure you understand it well and explain your choice in your assignment.

Q: Can I use external data for comparison?

A: You may reference external sources in your report for context, but your analysis and visualisations must be based on the provided dataset.

USE OF RESOURCES AND TECHNOLOGIES INCLUDING GENERATIVE ARTIFICIAL INTELLIGENCE

For this assessment, students are permitted to use generative artificial intelligence tools (GAITs e.g., ChatGPT) to:

- clarify concepts, theories, ideas, etc., discussed in class
- generate preliminary ideas for writing and coding
- edit a working draft of the assessment
- read and summarise research and supporting evidence for the assessment

Students are **not** permitted to use GAITs to

- generate definitions or writing used in their final submission.
- produce counter-arguments or refine thinking on their final submission
- generate complete R code in their final submission.

Any of these actions will constitute and be treated as a breach of academic integrity.

LATE SUBMISSION

A maximum penalty of five percentage points of the total possible marks will be applied per day to late submissions, for up to a maximum of seven calendar days. Tasks that have not been submitted within the maximum number of additional late days will receive a mark of zero, unless otherwise specified in the late penalties section of the Unit Guide. Late submission for a task will only be permitted when specified in the unit guide. This provision does not apply to online exams or other assessment with a time-limit of less than 24 hours.

Where an application for Special Consideration is approved and the outcome is an extension to the due date of a task, submissions that are received after the new due date will be subject to late penalties that are calculated from the new due date. This only applies where the outcome is an extension to the due date – see the Special Consideration Policy for a schedule of all possible outcomes.